

# Transshipment: optimality conditions

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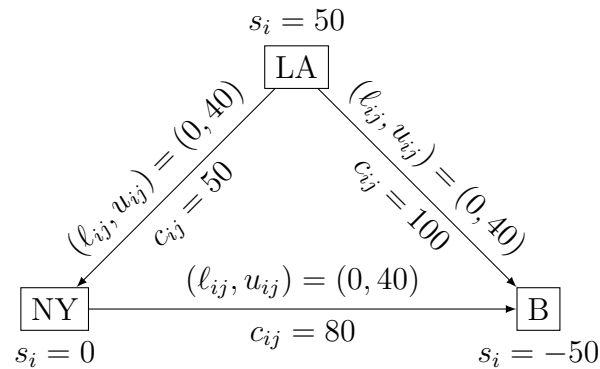
*Practice quiz*

Consider three factories producing electric cars in Boston, New-York and Los Angeles, with supplies 200 units, 250 units and 300 units per day respectively. The demand is 250 cars per day and per city. The cost for transporting one car:

- from Los Angeles to Boston is \$100.
- from Los Angeles to New-York is \$50.
- from New-York to Boston is \$80.

No more than 40 cars can be transported between two cities per day.

The corresponding network is:



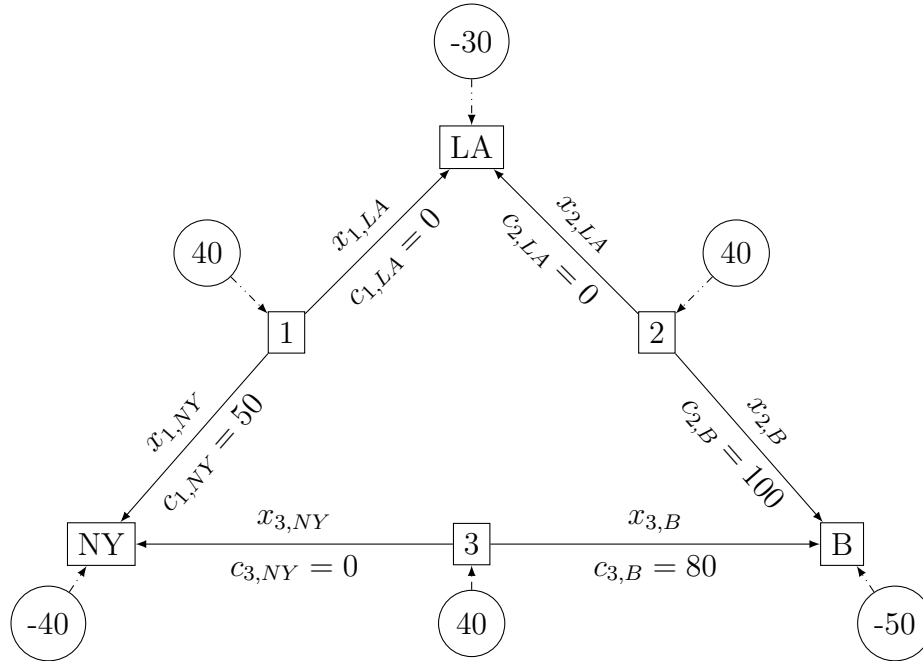
It corresponds to the following transshipment problem:

$$\min_{x \in \mathbb{R}^3} 100x_{LA,B} + 50x_{LA,NY} + 80x_{NY,B}$$

subject to

$$\begin{aligned}x_{LA,NY} + x_{LA,B} &= 50, \\x_{NY,B} - x_{LA,NY} &= 0, \\-x_{NY,B} - x_{LA,B} &= -50, \\0 \leq x_{LA,NY}, x_{LA,B}, x_{NY,B} &\leq 40.\end{aligned}$$

In order to obtain a formulation in standard form, we redesign the network in the following way:



The corresponding linear optimization problem in standard form is:

$$\min 100x_{2,B} + 80x_{3,B} + 50x_{1,NY}$$

subject to

$$-x_{1,LA} - x_{2,LA} = -30,$$

$$x_{1,LA} + x_{1,NY} = 40,$$

$$-x_{1,NY} - x_{3,NY} = -40,$$

$$x_{3,NY} + x_{3,B} = 40,$$

$$-x_{2,B} - x_{3,B} = -50,$$

$$x_{2,LA} + x_{2,B} = 40,$$

$$x_{1,LA}, x_{1,NY}, x_{2,LA}, x_{2,B}, x_{3,NY}, x_{3,B} \geq 0.$$

1. Write the dual problem using the Lagrangian method.
2. Solve the dual using optimality conditions.